

45. Cystic Kidney Diseases · assets 1-9 / 42

Stage	Cyst Wall	Septa	Calcified	Enhanced?	Work Up	Ultrasound	CECT	MRI
1	Hairline Thin Wall	No	No	No	No FU Needed			
2	Minimal Regular Wall Thickening	Thin, Smooth, Few Septa	Minimal	No	No FU Needed			
2F	Minimal Regular Wall Thickening	Multiple Minimal Smooth Thickening	Thick Nodular	No	FU at 6 months to 1 year, then annually for 5 years			
3	Thick Irregular on smooth Wall	Thick and Irregular	Thick Nodular and Irregular	Yes	Partial Nephrectomy or Radical Nephrectomy if not surgical candidates			
4	Irregular Gross Thickening	Irregular Gross Thickening	Thick Nodular and Irregular	Yes (Tissue and Cyst)	Partial or Total Nephrectomy			

Fig. 45.1 Characteristic findings on various abdominal imaging modalities and the criteria for the Bosniak Classification. CECT, Contrast-enhanced computed tomography scan; F/U, follow-up.

Fig_45_1

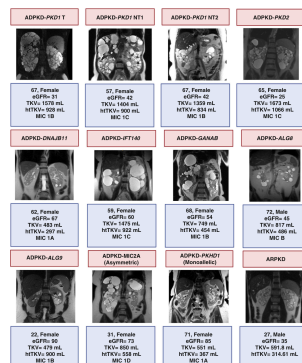


Fig. 45.4 Representative abdominal images of patients with autosomal dominant polycystic kidney disease seen on the large disease spectrum. GFR, Glomerular filtration rate; Ht, height; MAC, Mayo Imaging Classification; NT1, nontruncating tier 1; NT2, nontruncating tier 2; T, truncating; TKV, total kidney volume.

Fig_45_4

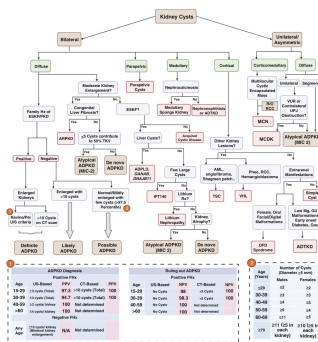


Fig. 45.7 Algorithm involved in diagnosing autosomal dominant polycystic kidney disease (ADPKD) based on the location of the kidney cysts, family history, extrarenal manifestations, and congruency between cystic burden and kidney function. Among the large differential, we emphasize how to diagnose definite, likely, and possible ADPKD. ADPLD, Autosomal dominant polycystic liver disease; ADPKD, autosomal dominant polycystic kidney disease; AML, arylsulphatase B deficiency; ESKD, end-stage kidney disease; H, history; MCKD, multicystic degenerative kidneys; MCKD, multicystic cystic nephrosis; MR, Mayo imaging classification; NPV, negative predictive value; OFD, oral facial digital; PPS, pheochromocytoma; PPV, positive predictive value; RCC, renal cell carcinoma; R, treatment; TKV, total kidney volume; TSC, tuberous sclerosis complex; URT, uropoietic function; VHL, von Hippel Lindau syndrome; VHL, von Hippel Lindau syndrome.

Fig_45_7

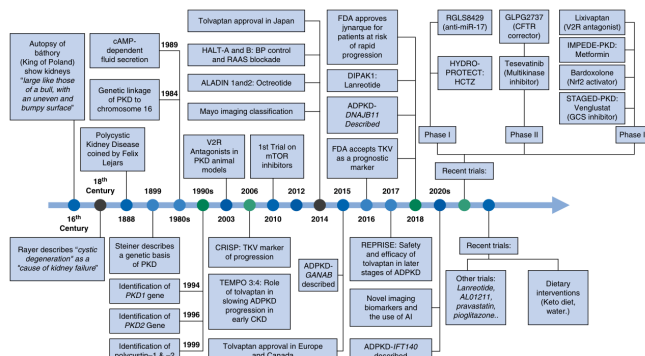


Fig. 45.2 Timeline of the major historical milestones and trials related to autosomal dominant polycystic kidney disease.

Fig_45_2

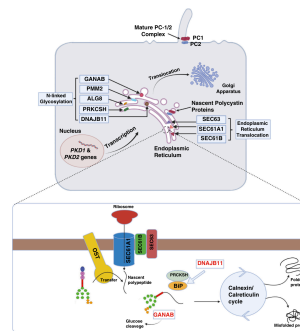


Fig. 45.5 Maturation process in the tubular epithelium. The maturation process of ACP/PC2-associated proteins, primarily polycystin 1 (PC1) and polycystin 2 (PC2), from gene expression to functional integration within the primary cilium. Initiated within the nucleus, genes encode PC1 and PC2 under transcription, producing mRNA transcripts. Subsequently, these transcripts are transported to the endoplasmic reticulum (ER), where translation takes place, leading to the synthesis of PC1 and PC2 polypeptides. Crucial proteins in this maturation cascade, including *GRASP48* (involved in ER translocation) and *GNAI1*, *AGG*, *QSOX1*, *PMAD*, and *PRKCSH* (involved in N-linked glycosylation), are important roles in guiding proper folding, post-translational modifications, and trafficking of PC1 and PC2. Notably, *DNAH17* emerges as a facilitator in ensuring the maturation and correct trafficking of PKD1. Additionally, the figure underscores the significance of various proteins in the accurate placement of PC1 and PC2 within the ciliary structure, where their functional roles are realized.

Fig_45_5

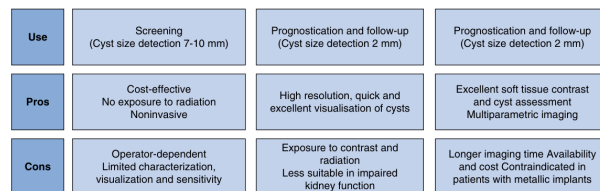


Fig. 45.8 Abdominal Imaging representation of a 64-year-old man with autosomal dominant polycystic kidney disease. (A) Ultrasound. (B) Computed tomography scan transverse section. (C) Magnetic resonance imaging coronal section.

Fig_45_8

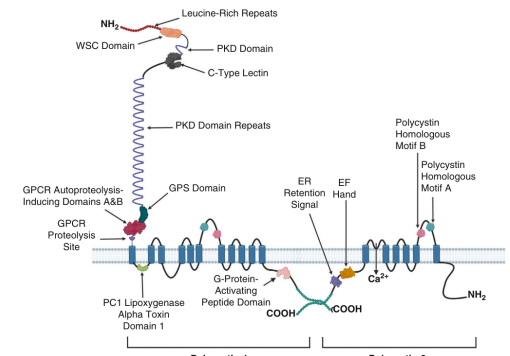
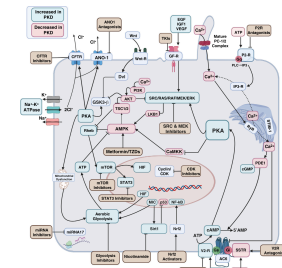


Fig. 45.3 Structure of polycystin proteins. Polycystin 1 (PC1) and polycystin 2 (PC2) form the polycystin complex.

Fig_45_3

[illegible]

Fig_45_6

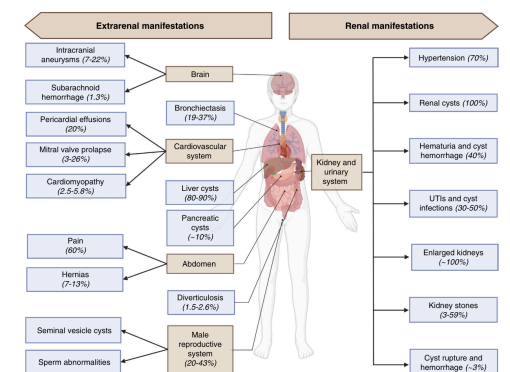


Fig. 45.9 Summary of the renal and extrarenal manifestations seen in autosomal dominant polycystic kidney disease.

Fig_45_9

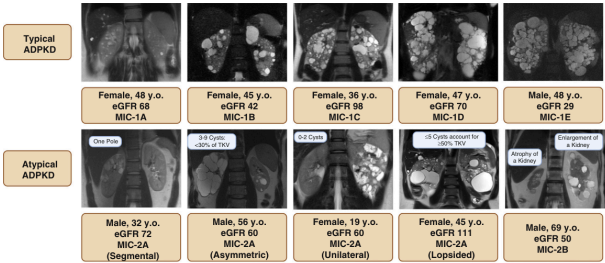


Fig. 45.10 Magnetic resonance imaging representation of typical (MIC-1A, 1B, 1C, 1D, and 1E) versus atypical (MIC-2A and 2B) autosomal dominant polycystic kidney disease.

Fig_45_10

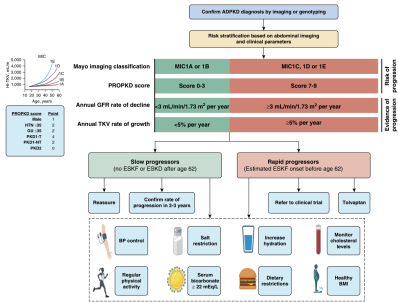


Fig. 45.13 Treatment diagram of patients with autosomal dominant polycystic kidney disease (ADPKD). ADPKD management starts with a confirmation of the diagnosis through imaging or genotyping. Subsequently, patients are stratified based on abdominal imaging and clinical parameters. Slow progressors, classified as MIC-1A or 1B with a PROPKD score of 0 to 3, an annual GFR rate of decline <3 mL/min/1.73 m² per year, and an annual TEV rate of growth $<5\%$ per year, are identified. These individuals are expected to either not develop and stage kidney disease (ESKD) or do so after the age of 62. Conversely, rapid progressors, categorized as MIC-1C, 1D, or 1E with a PROPKD score of 7 to 9, an EGRF rate of decline >3 mL/min/1.73 m² per year, and a TEV rate of growth $>5\%$ per year, require prompt intervention as ESKD is estimated before the age of 62. Management strategies differ accordingly. Slow progressors are monitored and their progression is recommended in 2 to 3 years, while rapid progressors are referred to clinical trials and may be listed on transplant. Additionally, all patients are advised to adhere to lifestyle modifications including blood pressure control, regular physical activity, salt restriction, maintenance of serum bicarbonate levels >22 , increased hydration, dietary adjustments, monitoring of cholesterol levels, and maintenance of a healthy body mass index (BMI).

Fig_45_13



Fig. 45.16 Imaging representation of a 2-year-old male patient with TSC2/PD1 continuous gene syndrome (eGFR at imaging of 108 mL/min/1.73 m²).

Fig_45_16

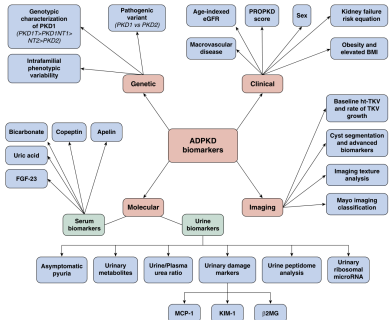


Fig. 45.11 The different biomarkers identified in autosomal dominant polycystic kidney disease (ADPKD) progression and prognosis. This detailed classification organizes ADPKD biomarkers into four distinct groups. Genetic biomarkers include the type of genetic variant, highlighting the role of PKD1 pathogenic variants in disease presentation. Further delineation refers PKD1 pathogenic variants to macrovascular disease and intrafamilial variations. Clinical markers including age-related eGFR, PROPKD score, and considerations of macrovascular disease provide insights into risk factors, with males experiencing more severe ADPKD. Imaging biomarkers such as TEV and TEV growth rate are also proposed as prognostic biomarkers in ADPKD while advanced imaging techniques offer nuanced insights. Molecular markers, divided into serum and urinary categories, include apelin, copeptin, bicarbonate, uric acid, FGF23, asymptomatic pyuria, urinary/serum urea ratio, and more, offering systemic and renal-specific perspectives.

Fig_45_11

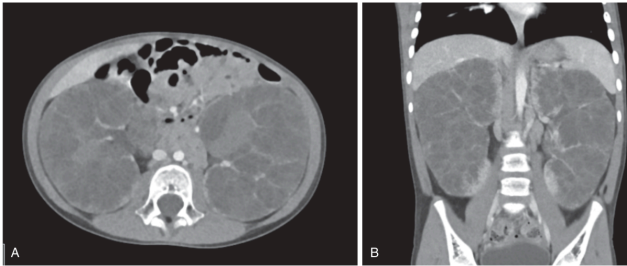


Fig. 45.14 Computed tomography scans (A, cross-sectional cut vs. B, coronal cut) of a 5-year-old female patient with NEK8 variant and eGFR of 25 mL/min/1.73 m², manifesting with early onset autosomal dominant polycystic kidney disease.

Fig_45_14

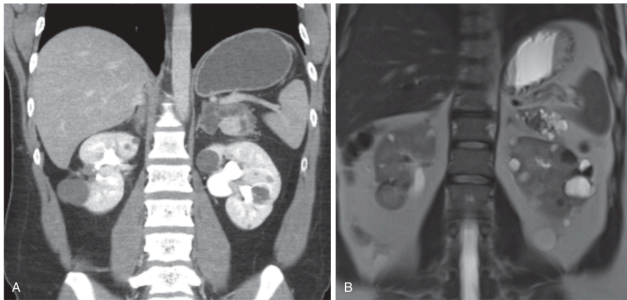


Fig. 45.17 Imaging representation of a 38-year-old male with Von Hippel-Lindau disease on computed tomography scan (A) and magnetic resonance imaging (B) (estimated glomerular filtration rate at imaging of 85 mL/min/1.73 m²).

Fig_45_17

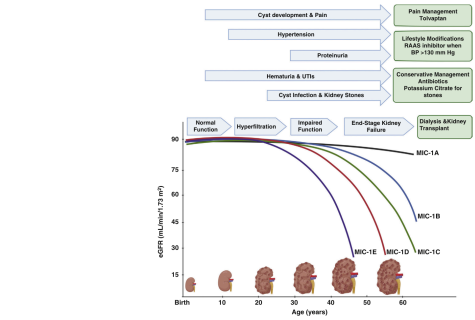


Fig. 45.12 Disease progression in autosomal dominant polycystic kidney disease (ADPKD). This representation outlines the progressive stages of ADPKD in the context of chronic kidney disease (CKD). Starting with normal kidney function, the trajectory advances through hypertension and impaired function, ultimately leading to end-stage kidney failure (ESKF), necessitating interventions like dialysis or transplantation. Noteworthy early manifestations include cyst development, pain, hematuria, urinary tract infections (UTIs), and hypertension, with later stages featuring proteinuria, cyst infections, and stone formation. Different interventions are shown for different manifestations: telaprost for limiting cystic and disease progression, renin-angiotensin-aldosterone system (RAAS) blockers for hypertension, potassium citrate for stones, and antibiotics for UTIs. The comprehensive approach of conservative management proves crucial in addressing various ADPKD-associated symptoms including hematuria, stones, and pain, contributing to comprehensive patient care. MIC, Mayo Imaging Classification.

Fig_45_12

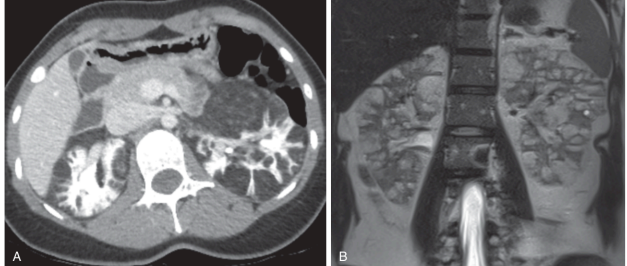


Fig. 45.15 Imaging representation of a 33-year-old woman diagnosed with tuberous sclerosis complex complex (eGFR at imaging of 60 mL/min/1.73 m²) on computed tomography scan (A) and magnetic resonance imaging (B) (arrow showing angiomyolipomas).

Fig_45_15



Fig. 45.18 Magnetic resonance imaging of a 32-year-old woman with HNF1B-associated kidney disease (estimated glomerular filtration rate at imaging of 90 mL/min/1.73 m²).

Fig_45_18



Fig. 45.19 Magnetic resonance imaging of a 39-year-old woman with oral-facial-digital syndrome type 1 (estimated glomerular filtration rate at imaging of 52 mL/min/1.73 m²).

Fig_45_19

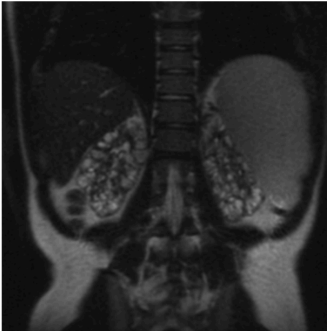


Fig. 45.20 Magnetic resonance imaging of a 22-year-old woman with autosomal recessive polycystic kidney disease (estimated glomerular filtration rate at imaging of 16 mL/min/1.73 m²).

Fig_45_20

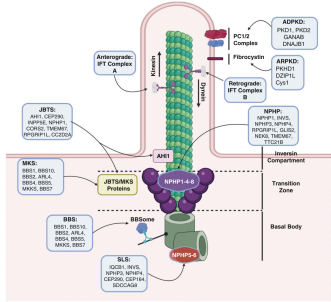


Fig. 45.21 Structure of the primary cilium including the distribution of cilioproteins and the disease and genes implicated in the ciliary disease. This diagram illustrates the structure of the primary cilium, a sensory organelle implicated in the pathogenesis of polycystic kidney disease (PKD) and various ciliopathies. The primary cilium houses distinct groups of cilioproteins, and multiple transport (PT) mechanisms play a crucial role in protein trafficking within the cilium. Two specific complexes, PT Complex 1 and PT Complex 2, both the lowest 2 motor for anterograde movement and PT Complex 3, with the cilium motor for retrograde movement, contribute to cilia formation. Proteins associated with Nephron and Joubert syndrome are involved in these transport processes. Nephron syndrome proteins (NPHP1, NPHP2, NPHP3, NPHP4, NPHP5, NPHP6, NPHP7, NPHP8, NPHP9, NPHP10, NPHP11, NPHP12, NPHP13, NPHP14, NPHP15, NPHP16, NPHP17, NPHP18, NPHP19, NPHP20, NPHP21, NPHP22, NPHP23, NPHP24, NPHP25, NPHP26, NPHP27, NPHP28, NPHP29, NPHP30, NPHP31, NPHP32, NPHP33, NPHP34, NPHP35, NPHP36, NPHP37, NPHP38, NPHP39, NPHP40, NPHP41, NPHP42, NPHP43, NPHP44, NPHP45, NPHP46, NPHP47, NPHP48, NPHP49, NPHP50, NPHP51, NPHP52, NPHP53, NPHP54, NPHP55, NPHP56, NPHP57, NPHP58, NPHP59, NPHP60, NPHP61, NPHP62, NPHP63, NPHP64, NPHP65, NPHP66, NPHP67, NPHP68, NPHP69, NPHP70, NPHP71, NPHP72, NPHP73, NPHP74, NPHP75, NPHP76, NPHP77, NPHP78, NPHP79, NPHP80, NPHP81, NPHP82, NPHP83, NPHP84, NPHP85, NPHP86, NPHP87, NPHP88, NPHP89, NPHP90, NPHP91, NPHP92, NPHP93, NPHP94, NPHP95, NPHP96, NPHP97, NPHP98, NPHP99, NPHP100) are involved in the transition zone, regulating cilium protein composition. Specific proteins like NPHP1, NPHP2, NPHP3, NPHP4, NPHP5, NPHP6, NPHP7, NPHP8, NPHP9, NPHP10, NPHP11, NPHP12, NPHP13, NPHP14, NPHP15, NPHP16, NPHP17, NPHP18, NPHP19, NPHP20, NPHP21, NPHP22, NPHP23, NPHP24, NPHP25, NPHP26, NPHP27, NPHP28, NPHP29, NPHP30, NPHP31, NPHP32, NPHP33, NPHP34, NPHP35, NPHP36, NPHP37, NPHP38, NPHP39, NPHP40, NPHP41, NPHP42, NPHP43, NPHP44, NPHP45, NPHP46, NPHP47, NPHP48, NPHP49, NPHP50, NPHP51, NPHP52, NPHP53, NPHP54, NPHP55, NPHP56, NPHP57, NPHP58, NPHP59, NPHP60, NPHP61, NPHP62, NPHP63, NPHP64, NPHP65, NPHP66, NPHP67, NPHP68, NPHP69, NPHP70, NPHP71, NPHP72, NPHP73, NPHP74, NPHP75, NPHP76, NPHP77, NPHP78, NPHP79, NPHP80, NPHP81, NPHP82, NPHP83, NPHP84, NPHP85, NPHP86, NPHP87, NPHP88, NPHP89, NPHP90, NPHP91, NPHP92, NPHP93, NPHP94, NPHP95, NPHP96, NPHP97, NPHP98, NPHP99, NPHP100) are involved in the basal body, regulating cilium protein composition. Variants in any of these proteins disrupt the normal composition, sensory functions, and signaling of the cilium, leading to pleiotropic phenotypic characteristics of ciliopathies. Understanding these molecular encodes provides insights into the pathogenesis of diseases associated with primary cilia dysfunction.

Fig_45_21

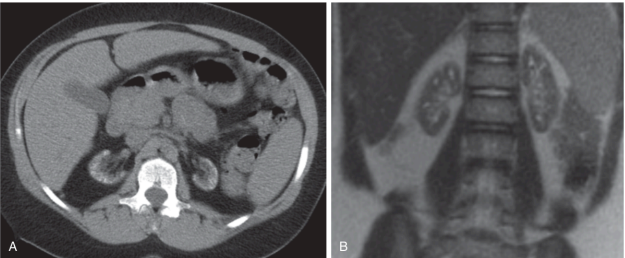


Fig. 45.22 Imaging representation of female patient with Joubert syndrome. (A) Computed tomography scan at 13 years of age with eGFR of 54.7 mL/min/1.73m². (B) Magnetic resonance imaging at 10 years of age with eGFR of 77.6 mL/min/1.73 m².

Fig_45_22

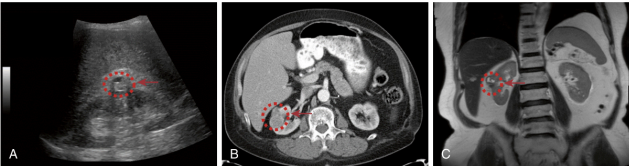


Fig. 45.23 Imaging representation of renal cell carcinoma in a 75-year-old male patient (eGFR = 49 mL/min/1.73 m²) on ultrasound (A), transverse computed tomography (B), and coronal magnetic resonance imaging (C).

Fig_45_23

Table 45.1 Differences Between Simple and Complex Cysts		
	Simple Cyst	Complex Cyst
Characteristics	Most common renal lesion containing water-like fluid	Less common renal mass with thicker walls and solid-like material
Presentation	Typically, asymptomatic. Dull abdominal or back pain and urinary changes in case of enlargement	Typically, asymptomatic. May present with fever, pain, and hematuria
Imaging	Incidental finding. Oval anechoic mass	Poorly demarcated thick cyst walls with septations and internal echoes reflecting solid-like material inside.

Table 45.2 Comparison Between Autosomal Dominant Polycystic Kidney Disease (ADPKD), Multiple Simple Cysts, Unilateral Renal Cystic Disease (URCD), Multicystic Dysplastic Kidneys (MCDK), and Multiple Unilateral Subcapsular Hemorrhagic (MUCH)			
Disease	Acquired vs. Inherited	Prevalence	Clinical Features
ADPKD	Typical (MIC 1A-1E) Atypical (MIC 2A-2B)	Autosomal dominant 1/1000-1/2500	Bilateral cyst involvement with extrarenal manifestations. Leads to several complications including cyst hemorrhage, hypertension, and cerebral aneurysms Focal disease (MIC-2A) or patterns of parenchymal atrophy (MIC-2B). Milder form of the disease with a later onset of symptoms, slower cyst growth, and lower risk of complications
Multiple Simple Cysts	Acquired	Common	Usually unilateral but may increase and show up independently in both kidneys with age.
Unilateral Renal Cystic Disease (URCD)	Acquired	Unknown	Typically, asymptomatic. Looks like PKD but manifests unilaterally. Does not typically progress to ESKF and lacks extrarenal manifestations and complications.
Multicystic Dysplastic Kidney disease (MCDK)	Acquired	1/1000-1/4300	Group of cysts of different sizes. May lead to complications such as vesicoureteral reflux.
Multiple Unilateral Subcapsular Hemorrhagic (MUCH) Cysts	Acquired	Unknown	Multiple unilateral small kidney cysts displaying hemorrhagic contents (hyper-attenuated on unenhanced CT or extremely hypointense on T2-weighted magnetic resonance image)
ESKF, End-stage kidney failure; MIC, Mayo imaging classification.			

Table_45_2

Table 45.3 Table Representing National and International Prevalence of ADPKD			
Location	Year	Prevalence	
Global	1997	Point prevalence ranges from 1 in 400 to 1 in 1000 live births, affecting approximately 12.5 million people globally. ¹¹	
	2017	Lifetime prevalence of 5.3 per 10,000, out of which PKD1 constitutes 6.8 per 10,000 and PKD2 2.4 per 10,000. HNF1B truncating variant detected in 1 per 28,770 individuals while TSC complex 1 in 2310 individuals. Minimal genetic prevalence 9.3 per 10,000 individuals. ¹⁸	
European Union	2012	Screening prevalence 3.08 in 10,000 Minimum point prevalence 3.29 in 10,000 51.1 million on RRT ¹⁹	
United States	2002-2018	Overall crude prevalence 4.26 per 10,000. When standardized to age, prevalence decreases to 4.15 per 10,000. ¹¹	
	1835-1980	ADPKD prevalence of 100-250/100,000 inhabitants ¹²	
	1980-2016	2010 Prevalence: Definite ADPKD: 4.7 per 10,000 Possible ADPKD: 12.4 per 10,000 2010 incidence: Definite ADPKD: 179 per 100,000 Definite, likely, or possible ADPKD: 8.44 per 100,000 ¹¹	
	2013-2015	Commercial and Medicare Database 2013 Prevalence: 1.74 per 10,000 2014 Prevalence: 1.87 per 10,000 2015 Prevalence: 2.10 per 10,000 Managed Medicaid Database: 2013 prevalence: 2.26 per 10,000 2014 prevalence: 2.40 per 10,000 2015 prevalence: 2.20 per 10,000 ¹¹	
	2016-2017	2016 prevalence: 2.34 per 10,000 2-yr prevalence: 3.61 per 10,000 ¹¹	
Seychelles	1993-1995	3-yr prevalence 5.7 per 10,000 1 in 54 among the Caputian population ¹¹	
United Kingdom	1991	Prevalence: 1 in 2450 ²⁰	
France	1996	Minimum prevalence 3.9 per 10,000 and screening prevalence: 4.6 ¹⁸	
Germany	1999	Minimum prevalence 2.4 per 10,000, screening prevalence 3.3 ¹⁸	
	2013	overall prevalence of ADPKD in southeast Germany was 3.27/10,000; ranging from 1.73/10,000 in the third decade to 5.73/100 000 in the 8th decade of life ²¹	
Modena, Italy	1980-2017	Point prevalence: 3.0 per 10,000 and predicted prevalence 4.76 per 10,000 ¹¹	

Table_45_3

Table 45.4 Main Differences Among ADPKD-Spectrum Genes, Expression, and Manifestations										
Gene	Disease	Chr.	Year	MD	Protein	Protein Function	Proportion Manifestations	Kidney	Renal	Liver
PKD1	ADPKD-PKD1	16p13.3	1984	AD	Polycystin-1	Receptor. Complex with PC2. 70% of ADPKD cases have well understood	70%	Multiple bilateral cysts. Highest cystic burden among other variants	ESKF at mean age 56.1	Absent to severe
PC2	ADPKD-PC2	4q21	1986	AD	Polycystin-2	Cx43-permeable nonselective cation channel. Complex with PC1	10%	Multiple bilateral cysts	ESKF at mean age 75.7	Absent to severe
IFT140	ADPKD-IFT140	16p13.3	2021	AD	IFT140 Protein (part of IFT complex-A)	Retinopodia ciliary transport. Well developed and functioning of cilia	1-2%	Few, large bilateral cysts	Preserved GFR until old age	Rare
GNA01B	ADPKD-GNA01B	11q12.3	2016	AD	Subunit of G-protein-coupled receptor-11	Catalytic subunit of G-protein-coupled receptor-11	<0.2%	Highly variable. Mild cystic burden	Limited CKD, no ESKF	Mild to severe
DNAH11	ADPKD-DNAH11	9q27.3	2018	AD	DNAH11 (part of Dyx11 complex)	Regulation of cilia and cell surface protein. Modulation and correct trafficking of PKD1	<0.5%	Bilateral, small cysts, mild cystic burden	Limited early CKD	Mild
ALG3	ADPKD-ALG3	16p13.3	2022	AD	Aldehyde dehydrogenase 3	Assembly of oligosaccharides in kidney epithelial cells. Proper PKD1 glycosylation and trafficking	<0.2%	Mild. Moderate cystic burden, mild oligosaccharidosis	Preserved GFR until old age	Few, rarely
ALG8	ADPKD-ALG8	10p13.3	-	AD	1-3-glycosyltransferase	Modification and localization of PKD1 into the primary cilia	<0.2%	Mild cystic burden	Generally preserved GFR	Liver cysts are present with PKD
ALG8	ADPKD-ALG8	11q14.1	2017	AD	1-3-glycosyltransferase	Modification and localization of PKD1 into the primary cilia	<1%	Mild cystic burden	Preserved GFR until old age	Mild to severe
ALG8	ADPKD-ALG8	11q14.1	2019	AD	1-3-glycosyltransferase	Modification and localization of PKD1 into the primary cilia	<0.2%	Mild to moderate cystic burden	Preserved GFR until old age	Common
PKD2	ADPKD-PKD2	6p12.3	1987	AD	Polycystin-2	Protein PKD2 mutation. Cx43-Cx36 and Cx43-Cx36	<1%	Highly variable. Mild cystic burden	Preserved GFR until old age	Common

Chr, Chromosome; CKD, chronic kidney disease; ESKF, end-stage kidney failure; ICA, intracranial aneurysms; Manifest, manifestation; MD, mode of inheritance; Prop, proportion.

Table_45_4

Table 45.5 Performance of Ultrasound-Based Unified Criteria for Diagnosis or Exclusion of ADPKD in Patients (*PKD1* or *PKD2* pathogenic variants) With Positive Family History

Age (yr)	Imaging Findings	<i>PKD1</i>	<i>PKD2</i>	Unknown Gene Type
For Diagnostic Confirmation				
15-29	A total of ≥3 cysts*	PPV = 100% Sensitivity = 94.3% PPV = 100%	PPV = 100% Sensitivity = 69.5% PPV = 100%	PPV = 100% Sensitivity = 81.7% PPV = 100%
30-39	A total of ≥3 cysts*	Sensitivity = 96.6% PPV = 100%	Sensitivity = 94.9% PPV = 100%	Sensitivity = 95.5% PPV = 100%
40-59	≥2 cysts in each kidney	PPV = 100% Sensitivity = 92.6%	PPV = 100% Sensitivity = 88.8%	PPV = 100% Sensitivity = 90%
For Diagnostic Exclusion				
15-29	No kidney cyst	NPV = 99.1% Specificity = 97.6%	NPV = 80.5% Specificity = 96.6%	NPV = 90.8% Specificity = 97.1%
30-39	No kidney cyst	NPV = 100% Specificity = 96.0%	NPV = 96.8% Specificity = 93.8%	NPV = 98.3% Specificity = 94.8%
40-59	No kidney cyst	NPV = 100% Specificity = 93.9%	NPV = 100% Specificity = 93.7%	NPV = 100% Specificity = 93.9%

ADPKD, Autosomal dominant polycystic kidney disease; PPV, positive predictive value; NPV, negative predictive value; *, unilateral or bilateral. Adapted from Chapman AB, Devuyt O, Eckardt KU, et al. Autosomal-dominant kidney disease: executive summary from a kidney conference. *Kidney Int.* 2015;88:17–27.

Table_45_5

Table 45.6 Specific Values Above 97.5th Percentile by Age and Sex for Possible ADPKD Diagnosis; Cysts Must Be ≥5 mm in Diameter

Age (Years)	Number of Cysts (Diameter ≥5 mm)	
	Males	Females
≤29	≥2	≥2
30-39	≥3	≥3
40-49	≥4	≥3
50-59	≥6	≥4
60-69	≥11	≥5
≥70	≥11 (≥5 in each kidney)	≥10 (≥5 in each kidney)

Table_45_6

Table 45.8 Strength and Limitations of Various Methods Used to Assess Rapid Progression Risk in ADPKD²⁰⁰²

Prognostic Biomarker	Strengths	Limitations
eGFR (index by age)	Classic biomarker indicating rapid progression if congruent with cystic burden	Delay therapy in early stages. Does not exclude other factors affecting GFR
eGFR rate of decline	Measures actual rather than predicted progression	Delay therapy in young patients. Does not exclude non-ADPKD factors
htTKV (>600 mL/m)	Approved predictor of CKD 3	CT/MRI are costly, and US is less precise. Unreliable in atypical ADPKD
TKV rate of growth (>5%/yr)	Measures actual growth	Needs expertise for precise measurements and unreliable for atypical ADPKD
MIC (classes 1C, 1D, and 1E)	Predicts eGFR decline and practical for atypical ADPKD	Costly, small class separation at young age and underrepresents non-White population
Kidney length >16.5 cm in patients aged <45 yr	Less expensive, correlates with TKV >750 mL and predicts CKD stage 3	Denies treatment for some young patients, mislabel atypical ADPKD cases, and operator dependent
Genetic testing (<i>PKD1</i> variant)	Links variant to severity	High cost and limited availability. More accurate at the population rather than individual level
Family history	Predictive of <i>PKD1</i> variant	Variable accuracy and no individual precision
PROPKD (Score >6)	Combines clinical and genetic markers and high chance of reaching ESKF by age 60	Not helpful in patients <35 yr unless hypertensive with urologic complications

eGFR, Estimated glomerular filtration rate; ESKF, end-stage kidney failure; htTKV, height-adjusted TKV. Adapted from Chebib FT, Torres VE. Assessing risk of rapid progression in autosomal dominant polycystic kidney disease and special considerations for disease-modifying therapy. *AJKD.* 2021;78(2):282–292.

Table_45_8

Categories	Mechanism of Action	ADPKD Preclinical Studies	Clinical Trials
mTOR Inhibitors Everolimus Sirolimus	Inhibition of mTOR prevents PKC-dependent activation of PKD1, leading to decreased cell proliferation and increased cell apoptosis	An animal model of slowly progressing PKD1 mice (mTOR ^{+/+}) showed that AKAP2 (an mTOR inhibitor) reduced the growth of slowly progressing cysts ¹¹	The animal study ¹¹ did not show a significant impact on TKV or eGFR. However, a phase 2 trial showed that AKAP2 (an mTOR inhibitor) reduced the growth of slowly progressing cysts ¹¹
CTF5 Modulators mTOR GPR127	CTF5 modulates the mTOR pathway and has been shown to be involved in regulating cell growth. Studies show that it regulates the mTOR pathway	An animal model of slowly progressing PKD1 mice (mTOR ^{+/+}) showed that AKAP2 (an mTOR inhibitor) reduced the growth of slowly progressing cysts ¹¹	The animal study ¹¹ did not show a significant impact on TKV or eGFR. However, a phase 2 trial showed that AKAP2 (an mTOR inhibitor) reduced the growth of slowly progressing cysts ¹¹
Epigenetic Agents	Epigenetic agents regulate gene expression and have been shown to be involved in regulating cell growth	Epigenetic agents regulate gene expression and have been shown to be involved in regulating cell growth	Epigenetic agents regulate gene expression and have been shown to be involved in regulating cell growth
mTOR Inhibitors Everolimus Sirolimus	mTOR inhibitors regulate cell growth and have been shown to be involved in regulating cell growth	mTOR inhibitors regulate cell growth and have been shown to be involved in regulating cell growth	mTOR inhibitors regulate cell growth and have been shown to be involved in regulating cell growth
Timed-release mTOR	Timed-release mTOR inhibitors regulate cell growth and have been shown to be involved in regulating cell growth	Timed-release mTOR inhibitors regulate cell growth and have been shown to be involved in regulating cell growth	Timed-release mTOR inhibitors regulate cell growth and have been shown to be involved in regulating cell growth

Continued on following page

Table_45_9_part1

Diagnosis	Gene	Age of Onset	Imaging Findings	Differences With ADPKD
Simple Cysts	PKD1, PKD2, PKD3, PKD4, PKD5, PKD6, PKD7, PKD8, PKD9, PKD10, PKD11, PKD12, PKD13, PKD14, PKD15, PKD16, PKD17, PKD18, PKD19, PKD20, PKD21, PKD22, PKD23, PKD24, PKD25, PKD26, PKD27, PKD28, PKD29, PKD30, PKD31, PKD32, PKD33, PKD34, PKD35, PKD36, PKD37, PKD38, PKD39, PKD40, PKD41, PKD42, PKD43, PKD44, PKD45, PKD46, PKD47, PKD48, PKD49, PKD50, PKD51, PKD52, PKD53, PKD54, PKD55, PKD56, PKD57, PKD58, PKD59, PKD60, PKD61, PKD62, PKD63, PKD64, PKD65, PKD66, PKD67, PKD68, PKD69, PKD70, PKD71, PKD72, PKD73, PKD74, PKD75, PKD76, PKD77, PKD78, PKD79, PKD80, PKD81, PKD82, PKD83, PKD84, PKD85, PKD86, PKD87, PKD88, PKD89, PKD90, PKD91, PKD92, PKD93, PKD94, PKD95, PKD96, PKD97, PKD98, PKD99, PKD100, PKD101, PKD102, PKD103, PKD104, PKD105, PKD106, PKD107, PKD108, PKD109, PKD110, PKD111, PKD112, PKD113, PKD114, PKD115, PKD116, PKD117, PKD118, PKD119, PKD120, PKD121, PKD122, PKD123, PKD124, PKD125, PKD126, PKD127, PKD128, PKD129, PKD130, PKD131, PKD132, PKD133, PKD134, PKD135, PKD136, PKD137, 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45. Cystic Kidney Diseases · assets 37-42 / 42

Table 45.12 Comprehensive Overview of Diverse Kidney-Related Diseases: Etiology, Phenotypes, and Clinical Management						
Disease or Syndrome	Etiology	Prevalence	Age at Presentation	Kidney Phenotype	Extrarenal Manifestation	Prognosis/Management
A. Rare Diseases and Syndromes Associated with Kidney Cysts						
Haga-Cheney syndrome	AD, NOTCH2 gene variant. Can occur sporadically ¹⁰²	1/1,000,000 ¹⁰²	Infancy-Childhood Age-dependent progression ¹⁰²	Bilateral polycystic kidneys	Skeletal (Craniofacial and dental abnormalities, bone deformities, and osteoclastosis), neurological complications, congenital heart defects, mitral and aortic valve abnormalities, respiratory infections, hearing loss, sleep apnea ¹⁰²	Morbidity and mortality increased due to bony fractures. Nephroses prognosis dependent based on invagination can occur in 50% of patients leading to central respiratory arrest ¹⁰²
Alagille syndrome variant	AD, JAG1, NOTCH2 variant	1/30,000-70,000 live births ¹⁰³ (increased to 1/30,000 with molecular diagnosis) ¹⁰³	Early childhood	Renal dysplasia, renal tubular acidosis, vesicular ureteric reflux, urinary obstruction, ¹⁰⁴ small echogenic kidney and kidney cysts, ¹⁰⁵⁻¹⁰⁷ hypoplasia of the branch pulmonary arteries, tetralogy of Fallot, vesicular (perirenal blood), systemic vascular anomalies of the aorta, and skeletal (Butterfly vertebrae, increased risk for fracture), and ocular abnormalities ¹⁰⁸ .	Liver (cholestasis, bile duct paucity, pruritus, xanthomas), cardiovascular anomalies (hypoplasia of the branch pulmonary arteries, tetralogy of Fallot, vesicular (perirenal blood), systemic vascular anomalies of the aorta, and skeletal (Butterfly vertebrae, increased risk for fracture), and ocular abnormalities ¹⁰⁸).	Significant morbidity and mortality due to the prevalence of cardiovascular anomalies (8-year survival decrease to 25% compared with hypoplasia of the branch pulmonary arteries, tetralogy of Fallot, vesicular (perirenal blood), systemic vascular anomalies ¹⁰⁸).

Table_45_12_part1

Table 45.12 Comprehensive Overview of Diverse Kidney-Related Diseases: Etiology, Phenotypes, and Clinical Management (Continued)						
Disease or Syndrome	Etiology	Prevalence	Age at Presentation	Kidney Phenotype	Extrarenal Manifestation	Prognosis/Management
Hypernatremic Hypoglycemia and Polycystic Kidney Disease (HPKD)	AR, PM2	1/50,000 for hypernatremic hypoglycemia ¹⁰⁹ ; 1/1,200,000 for HPKD ¹¹⁰	Infancy, childhood	Enlarged kidneys with multiple bilateral cysts	Hypoglycemia, cerebellar dysplasia, developmental delay, hepatic cysts, and abnormal fat distribution ¹¹⁰	Variable disease severity, mostly based on case reports and genetic studies ¹¹⁰
B. Urinary Stone Diseases						
HCO3 ⁻ variants and kidney cysts	HCO3A1 ¹¹¹	AR (AC)	Infancy-Adulthood	Stone formation, cystic kidneys, primary hyperparathyroidism (PTH), higher urinary calcium excretion compared with PH and PTH ¹¹²	Stone formation, cystic kidneys	Variable disease severity, mostly based on case reports and genetic studies ¹¹¹⁻¹¹³
Hypophosphatemic rickets with hypercalcaemia	SLC34A3 ¹¹⁴			Stone formation, cystic kidneys	Rarely hepatic cysts, hypophosphatemia, along with hypercalcaemia, osteoporosis, rickets, elevated 2,5-(OH)2D ³ levels	
CYP24A1 deficiency and kidney cysts	CYP24A1 ¹¹⁵			Stone formation, cystic kidneys, hypercalcaemia	Rarely hepatic cysts, reduced 24-hydroxylase activity, increased serum 1,25-dihydroxycholecalciferol levels along with and without hypercalcaemia	
C. Renal Cystic Diseases of Nongenetic Origin						
Cystic diseases of the renal sinus ¹¹⁶⁻¹¹⁷	Sporadic	Up to 1.5% of autopsy cases	Variable	Parapelvic or peripelvic cysts, kidney stones and renal hypophosphatosis, UTIs	Hypertension, abdominal pain, or proteinuria	Unknown. Can be managed by retrograde intrarenal holmium-laser resection ¹¹⁸

Continued on following page

Table_45_12_part2

Table 45.12 Comprehensive Overview of Diverse Kidney-Related Diseases: Etiology, Phenotypes, and Clinical Management (Continued)						
Disease or Syndrome	Etiology	Prevalence	Age at Presentation	Kidney Phenotype	Extrarenal Manifestation	Prognosis/Management
Perirenal lymphangiomas (peripelvic lymphangiectasia)	Congenital disease (secondary to obstruction of regional lymph drainage during development)	Difficult to estimate	Children	Hematuria, flank pain, abdominal pain, deterioration of kidney function over time if large enough to compress the kidney tissue ¹¹⁹⁻¹²¹	Commonly in the cervical neck and axilla, but may also be present in the retroperitoneum, mediastinum, mesentery, omentum, colon, and pelvis ¹¹⁹⁻¹²¹	Incidental finding. For most asymptomatic patients, treatment may not be necessary. Less invasive interventions, such as percutaneous or laparoscopic aspiration and cyst marsupialization, should be considered for regional lymphangiomas. Nephrectomy may be needed in decompensated cases ¹¹⁹⁻¹²¹
Subcapsular and perirenal urinoses (uriferous pseudocysts)	Sporadic, develop secondary to obstruction, pregnancy, or iatrogenic		Variable	Encapsulated collections of extravasated urine in the kidney parenchyma, mainly the subcapsular or perirenal space (not true cyst) ¹²²	Can be complicated by abscesses, hydronephrosis, electrolyte imbalances, and rupture ¹²²⁻¹²⁴	Worse prognosis with upper urinary tract obstruction. Possible spontaneous resolution but uncommon ¹²²

Table_45_12_part3

Table 45.12 Comprehensive Overview of Diverse Kidney-Related Diseases: Etiology, Phenotypes, and Clinical Management (Continued)						
Disease or Syndrome	Etiology	Prevalence	Age at Presentation	Kidney Phenotype	Extrarenal Manifestation	Prognosis/Management
Peripelvic Cysts (Peripelvic Diverticuli or Peripelvic Cysts)		All age groups are affected with an incidence of 0.7% among adults ¹²⁵⁻¹²⁷		Unilateral cystic enlargement of kidney pelvis or calyces. Urinary stasis and can be complicated by stones (10-40% of patients), infection, and loss commonly translocated on		Mostly asymptomatic management. Surgery is considered only if conservative management of complications proves ineffective ¹²⁹
Renal hepatoblastic interlobular stenosis and Megalocystosis				Obstruction to urine flow distal to the calyx, often linked to distortions at the calyx, urinary tract infection or kidney pelvis ¹²⁸		
				Megalocystosis indicates a cystic renal mass with increased volume and number of calyces without obstruction of abnormal renal function ¹³⁰		
Disease or Syndrome	Etiology	Prevalence	Age at Presentation	Kidney Phenotype	Extrarenal Manifestation	Prognosis/Management
D. Kidney Cyst-Related Dysplastic Diseases						

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Table_45_12_part4

Table 45.12 Comprehensive Overview of Diverse Kidney-Related Diseases: Etiology, Phenotypes, and Clinical Management (Continued)						
Disease or Syndrome	Etiology	Prevalence	Age at Presentation	Kidney Phenotype	Extrarenal Manifestation	Prognosis/Management
Cystic dysplasia of the kidney	Isolated, bilateral, rare in utero, at the level of ureteric bud, mesenchyme invagination ¹³²		In utero, neonatal, childhood	Cystic kidney dysplasia (multicystic), renal pelvis or calyces, and presence of normal tissue. Unlike renal cystic kidney diseases, which feature normal or enlarged kidney, cystic kidney dysplasia results usually in small kidney ¹³³	It can occur as an isolated condition or alongside other urinary or genetic disorders such as hemolytic renal syndrome (linked to ETV6 variants) and renal cystic syndrome (linked by PKD variants, among others VACTERL, Coleseley ¹³⁴)	Depending on the severity, can lead to chronic kidney disease, kidney failure ¹³⁵
Multicystic (Polycystic) Kidney (MCKD)	Familial (PKD), CYP24A1, CYP24A3, CYP24A4, CYP24A5, CYP24A6, CYP24A7, CYP24A8, CYP24A9, CYP24A10, CYP24A11, CYP24A12, CYP24A13, CYP24A14, CYP24A15, CYP24A16, CYP24A17, CYP24A18, CYP24A19, CYP24A20, CYP24A21, CYP24A22, CYP24A23, CYP24A24, CYP24A25, CYP24A26, CYP24A27, CYP24A28, CYP24A29, CYP24A30, CYP24A31, CYP24A32, CYP24A33, CYP24A34, CYP24A35, CYP24A36, CYP24A37, CYP24A38, CYP24A39, CYP24A40, CYP24A41, CYP24A42, CYP24A43, CYP24A44, CYP24A45, CYP24A46, CYP24A47, CYP24A48, CYP24A49, CYP24A50, CYP24A51, CYP24A52, CYP24A53, CYP24A54, CYP24A55, CYP24A56, CYP24A57, CYP24A58, CYP24A59, CYP24A60, CYP24A61, CYP24A62, CYP24A63, CYP24A64, CYP24A65, CYP24A66, CYP24A67, CYP24A68, CYP24A69, CYP24A70, CYP24A71, CYP24A72, CYP24A73, CYP24A74, CYP24A75, CYP24A76, CYP24A77, CYP24A78, CYP24A79, CYP24A80, CYP24A81, CYP24A82, CYP24A83, CYP24A84, CYP24A85, CYP24A86, CYP24A87, CYP24A88, CYP24A89, CYP24A90, CYP24A91, CYP24A92, CYP24A93, CYP24A94, CYP24A95, CYP24A96, 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